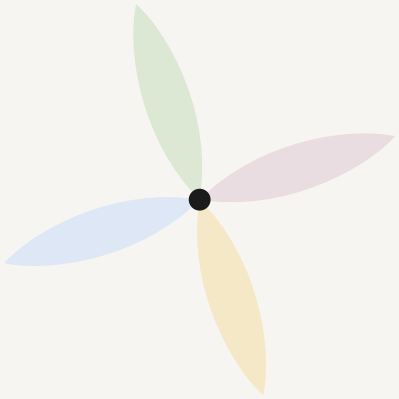
A PORTFOLIO OF ENGINEERING, EXPERIMENTS & SYSTEMS

Samanthula Eswar

Computer Science Engineering Student

I learn by building: research prototypes, useful software, and interfaces that make complex systems easier to understand.



Curiosity, made tangible.

Application portfolio / CMF Design Intern / 6 months

A study in form, tone and simple visual signals

Amrita Vishwa Vidyapeetham, Amritapuri / B.Tech CSE, 2027

GitHub Portfolio Email

A systems-minded builder

EDUCATION

B.Tech, Computer Science and Engineering

Amrita Vishwa Vidyapeetham, Amritapuri / Aug 2023 - expected 2027

Current CGPA: 7.72 (as listed on my résumé)

What I bring to a product team

Break systems down

Map inputs, constraints, states and failure cases before choosing an implementation.

Make ideas testable

Build prototypes and simulations that make an assumption visible and measurable.

Design for people

Use clear structure, responsive interfaces and feedback to make tools easier to use.

Technical toolkit

Programming	Python, Java, C/C++, JavaScript, TypeScript, SQL
Interfaces & services	React, Vite, Node.js, Express, Flask, HTML, CSS
Data & intelligent systems	PyTorch, OpenCV, NumPy, TIMM, scikit-learn, PostgreSQL, MongoDB, SQLite
Prototyping & delivery	Git/GitHub, Docker, MQTT, Node-RED, visual iteration through web projects

CMF is a new discipline for me. I am applying with a computer science background and a willingness to learn material, finish and colour methods from the ground up.

Trust, signal and identity

RESEARCH PROTOTYPE

THFSA – Hardware Fingerprint Identification

Explores whether image patterns can identify the fingerprint scanner that captured them. The prototype compares CNN, Transformer and hybrid models, then connects scanner verification to a Flask authentication demonstration.

Python / PyTorch / TIMM / EfficientFormer-L1 / Flask / SQLite / bcrypt / TOTP / HMAC-SHA256

Scope note: evaluated stored fingerprint images; no physical R307 scanner was connected. Dataset size and generalisation remain limitations.
Open repository

CAMERA AUTHENTICATION

PRNU Smartphone Authentication

A research implementation inspired by the ABC protocol. It trains camera sensor-noise fingerprints, challenges a phone through QR codes, and checks submitted images for device match, forgery and removal signals.

Pipeline: reference images > PRNU estimate > QR challenge > mobile capture/upload > PCE-based comparison and evaluation plots.

Python / OpenCV / NumPy / Flask / QR codes / PCE evaluation

Open repository

From signal to experience

IOT SYSTEM SIMULATION

Adaptive Traffic Signal Controller

A local simulation that adapts junction timing to demand. Sensor data is replayed through MQTT; Python analyses queues and selects bounded green times; a simulated actuator acknowledges the command; Node-RED and SQLite display state and history.

Design choice: one controller owns phase state; the dashboard observes and stores acknowledged state. This keeps intended actions distinct from what the simulation actually applied.

Python / MQTT / Mosquitto / Node-RED / SQLite / JavaScript

Open repository

FULL-STACK PRODUCT

Liftly - Ride-sharing coordination

A web application for drivers to publish trips and passengers to find and book rides. The client is built with React and TypeScript; an Express API and MongoDB support account, ride, booking, profile and messaging flows.

Product lens: reduce coordination friction by bringing route discovery, booking and communication into one flow. The portfolio describes repository features rather than claiming production scale.

React / TypeScript / Express / MongoDB / JWT

Open repository

Useful software, clearly shaped

StudyGenie AI

Turns subjects, deadlines and preferences into a personalised study plan. Built with React, TypeScript and Vite, with Gemini AI and Supabase integrations listed in the current project README. The interface makes a complex planning task legible through priorities and schedules.

React / TypeScript / Gemini AI / Supabase

Repository

Dormsy

A hostel management platform for students, wardens and administrators. Role-specific modules cover attendance, complaints, leave, fees and hostel operations. The work involves structuring many tasks into clear journeys and permissions.

React / TypeScript / Supabase / PostgreSQL

Repository

Design Pattern Case Study

An academic exploration of design patterns and architecture, using code analysis and UML / PlantUML diagrams. It strengthened my habit of explaining a system visually before or alongside implementation.

UML / PlantUML / NestJS / Java / Python

Repository

How I approach a new problem

Observe the context. Name the user need. Map constraints. Build a small test. Learn from the result. Refine the interaction and communicate the decision.

A curious start in CMF

I am a Computer Science Engineering student, not an Industrial Design graduate, and I have not worked professionally in CMF. I am drawn to this internship because product experience sits where technical performance meets what people see, touch and understand.

I would bring structured problem solving, prototyping habits and care for clear communication, while learning material, finish and colour selection from the team. I am especially interested in how small choices in surface, colour and interaction help a product feel intuitive and considered.

Let's connect

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github.com/eswarsamanthula

Portfolio - eswarsamanthula.github.io/Eswar.doodle-aesthetic/

APPLICATION DETAILS

Samanthula Eswar / Amrita Vishwa Vidyapeetham, Amritapuri / B.Tech CSE, expected 2027

Available for discussion about a 6-month, full-time, on-site internship in Hinjewadi, Pune; personal availability and relocation to be confirmed.